

Bilal Mustafa

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PROFILE SUMMARY

Seeking to leverage a strong quantitative background and 5+ years of client-facing experience, including 1.5+ years in healthcare workforce management and operations. Skilled in scheduling, forecasting, and collective agreement compliance, with proven ability to enhance operations using SQL, data visualization, and statistical modeling.

SKILLS

PROFESSIONAL SKILLS

Data Cleaning | Data Visualization | Statistical Analysis | Presentation Skills | Workforce Planning & Optimization | Collective Agreement Interpretation | Excellent Communication | Analytical Problem-Solving | ESP Training | ESP Timekeeping | Adult Learning Support

TECHNICAL SKILLS

SQL | T-SQL | Python | MS Excel | Google Sheets | MS Office | NumPy | Scikit Learn | Pandas | Matplotlib | Keras | TensorFlow | Folium | Seaborn | Dash | Google Suite | Tableau

WORK EXPERIENCE

Volunteer Data Analyst- CANAFRO RESEARCH INSTITUTE

Remote | JULY/2025-Present

- Analyzed Statistics Canada datasets on mental health, substance use, and socioeconomic trends among Black, African, and Caribbean Canadians.
- Conducted statistical tests (chi-square, residual analysis) to identify disparities and biases in mental health indicators.
- Designed and presented data visualizations (stacked bar charts, heat maps) for community research reports.
- Collaborated with the research team to standardize variables and ensure methodological rigor.

Staffing Clerk- STOLLERY NICU RAH

Alberta Health Services | Edmonton, AB | CAS May/2025-Current

- Communicated with stakeholders to ensure adequate staffing levels.
- Interpreted collective union agreements to ensure compliance and resolve payroll disputes.
- Fulfilled multiple ad-hoc scheduling requests placed by the unit manager through effective time management and collaborative team efforts.

Mental Health Referral System Analysis (Hackathon Project)

Converge Mental Health/ Data for Good | Edmonton, AB | Apr/2025

- Applied data analysis, natural language processing (NLP), and machine learning techniques using tools such as Python, Polars, Pandas, SQL, Tableau, Plotly, Sentence-BERT (SBERT), and Hugging Face Transformers.
- Designed semantic similarity models to evaluate referral quality between patient needs and services offered.
- Conducted exploratory data analysis (EDA), semantic similarity analysis, and statistical testing to identify service gaps and accessibility issues.

ADMINISTRATIVE ASSISTANT-IV (SCHEDULER)

Alberta Health Services | Edmonton, AB | (FT/CAS) Sep/2023-Current

- Communicated with stakeholders to ensure adequate staffing levels.
- Interpreted collective union agreements to ensure compliance and resolve payroll disputes.
- Fulfilled multiple ad-hoc scheduling requests placed by hospital units through effective time management and collaborative team efforts.
- Ensured accurate timekeeping and reporting using union guides and tableau dashboards.

OTHER

- Alberta Health Services | Edmonton, AB | 2022-2023
- Alberta Health Services- Public Health | Edmonton, AB | 2021-2022
- TD-Canada Trust | Edmonton, AB | 2017-2020

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PROJECTS

[fMRI AutoEncoder](#)

Personal Project | Applied neural networks to 3D MRI scans to explore cost-effective, AI-driven approaches for medical image analysis and reconstruction.

[Nursing Analysis](#)

Personal Project | Analyze and visualize data related to the nursing profession within Canada, data collected from various federal open data sources to create an interactive dashboard via Plotly Dash.

[Cancer CNN](#)

Personal Project | Utilize a CNN to analyze MRI scans of brain tumors to classify brain tumor type using Python with Tensorflow and Keras packages.

[Diabetes Analytics](#)

Personal Project | Analyze diabetes dataset obtained from 130 US hospitals from 1999-2008, to find trends and meaningful insights related to diabetes readmissions.

[Healthcare SQL](#)

Personal Project | Utilize SQL Server Management Service (SSMS) with T-SQL scripting to upload and analyze healthcare data to perform EDA and summarize data.

[Advanced SQL](#)

Personal Project | Utilize SSMS with T-SQL scripting to create a mock employee database and perform various basic and advanced analytical queries.

EDUCATION

Bachelor of Science -General (Physics Major/ Mathematics Minor)

2019

University of Alberta | Edmonton,AB

PROFESSIONAL DEVELOPMENT

IBM Data Science Certification

2022

Coursera | MOOC

Data Science Certification

2020

Northern Alberta Institute of Technology (NAIT) | Edmonton,AB